Appendices

Appendix A: Supplement to the ADP algorithm

A.1 The ADP implementation

The ADP algorithm replaces the expectation in Bellman equation (14) with a statistical estimate, resulting in an approximate Bellman equation as follows:

$$\max_{a_t \in A_t(S_t)} \left(\sum_{\delta \in [0,\upsilon]} R_{t+\delta}\left(s_{t+\delta}, a_{t+\delta}\right) + \bar{V}_{t+\upsilon}\left(s^a_{t+\upsilon}\right) \right) \tag{A1}$$

where $\bar{V}_t\left(s^a_t\right)$ denotes a piecewise linear function of post-decision state $s^a_t = \left(\tilde{y}_{i\tau t}, \tilde{\zeta}_{ht}\right)$. Then, the approximation accuracy is improved via repeated simulations. Specifically, offline learning detects the overarching MDP structure via batch learning. At the same time, the online case performs sequential updates to identify the local detailed structure.

For each offline simulation, a deterministic model with full realizations in episode ω^n is first constructed, which is then optimally solved by a node-based labeling algorithm, yielding the optimal solution, post-decision states $s^{n,a}_{t-1}$, and observation values $\hat{v}^n_t$. Then, the estimation will be smoothed as follows:

$$\bar{V}^n_{t-1}\left(s^{a,n}_{t-1}\right) = \left(1 - a_n\right)\bar{V}^{n-1}_{t-1}\left(s^{a,n}_{t-1}\right) + a_n \hat{v}^n_t \tag{A2}$$

where $\bar{V}^{n-1}_{t-1}\left(s^{a,n}_{t-1}\right)$ is the estimate of simulation $n-1$, and $a_n \in [0,1]$ denotes the step size.

In an online simulation, for a given period $t \in \mathcal{T}$, a dynamic model with short-term information and long-term approximations within the planning horizon $\left[t, \min\left(t+\upsilon, |\mathcal{T}|\right)\right]$ is solved to optimality using a period-based labeling algorithm, yielding the optimal decision a^n_t and observation value $\hat{v}^n_t$ of the current period. Then, the system state in the next period s^n_{t+1} is updated based on the current decision and the new realization. The procedure terminates upon returning to the destination. Similarly, the approximation is updated according to equation (A2) and improvement enhancements. Note that the step size a_n takes the average stepwise $a_n = 1/n$ and search-then-converge stepwise $\alpha_n = a_0\left(b/n + a\right)/\left(b/n + a + n^\beta\right)$ in offline and online learning, respectively.

A.2 The adaptive-interval method

The method starts with an initial partitioning $\hbar^k_{it}\left(k=0\right)$, which categorizes the colony strength $\mathcal{H}$ into $\theta_{\max}$ intervals of equivalent length. Each interval is indexed as

$\theta = \ell\left(\hbar_{it}^{k}, h\right)$, with its corresponding range $\left[h_{\theta}^{L}, h_{\theta}^{U}\right]$, accumulated frequency $\hat{\varpi}_{it\theta}^{k} = \sum_{h \in \left[h_{\theta}^{L}, h_{\theta}^{U}\right]} \varpi_{ith}^{n}$, and estimated coefficient $\lambda_{i\tau\theta t}^{n}$, where ϖ_{ith}^{n} denotes the frequency of visiting the state $s_{t}^{n,a}$ up to simulation n. Thus, the approximated value of a given post-decision state $s_{t}^{n,a} = \left(\tilde{y}_{i\tau t}^{n}, \tilde{\zeta}_{ht}^{n}\right)$ is denoted as

$$\bar{V}_{t}\left(s_{t}^{n,a}\right) = \sum_{i \in \mathcal{I}} \sum_{\tau \in \Gamma_{i}} \sum_{\theta \in \left[0, \theta_{\max}\right]} \sum_{h \in \left[h_{\theta}^{L}, h_{\theta}^{U}\right]} \lambda_{i\tau\theta t}^{n} \tilde{y}_{i\tau t}^{n} \tilde{\zeta}_{ht}^{n} \tag{A3}$$

As the process progresses, the frequency ϖ_{ith}^{n} increases by 1 upon each occurrence of state $s_{t}^{n,a} = \left(\tilde{y}_{i\tau t}^{n}, \tilde{\zeta}_{ht}^{n}\right)$. Every $K_{partition}$ iterations trigger the repartitioning procedure and yields a new partitioning $\hbar_{it}^{k+1}$ by updating interval boundaries, accumulated frequencies, and estimated coefficients. First, the upper bound h_{θ}^{U} of segment θ is determined if the cumulative frequency reaches θ times the average frequency, i.e., $\sum_{h \in \left[0, h_{\theta}^{U}\right]} \varpi_{ith}^{n} \geq \theta \sum_{h \in \mathcal{H}} \varpi_{ith}^{n} \Big/ \theta_{\max}$, or the number of remaining strength levels equals the number of unassigned segments, i.e., $\left|\mathcal{H}\right| - h \leq \theta_{\max} - \theta$; correspondingly, the left boundary turns $h_{\theta}^{L} = h_{\theta-1}^{U} + 1$. Then, the accumulated frequency and estimated coefficient are updated as $\hat{\varpi}_{it\theta}^{k+1} = \sum_{h \in \left[h_{\theta}^{L}, h_{\theta}^{U}\right]} \varpi_{ith}^{n}$ and $\lambda_{i\tau\theta t}^{n} = \sum_{h \in \left[h_{\theta}^{L}, h_{\theta}^{U}\right]} \varpi_{ith}^{n} \lambda_{i,\tau,\ell\left(\hbar_{it}^{k+1}, h\right),t}^{n} \Big/ \hat{\varpi}_{it\theta}^{k+1}$, respectively.

A.3 Pseudocodes for the LAMC and LATD algorithms

Algorithm A1 The LAMC framework

1: Initialization: Label set $\Psi_{o} = \{L\}$ at origin o, label set $\Psi_{i} = \varnothing$ at node $i \in \mathcal{I} \cup \{e\}$, handled nodes $\Lambda_{treated}$, set of nodes with untreated labels $\Lambda_{untreated} = \{o\}$, and set of untreated labels $UL = \varnothing$

2: While $\Lambda_{untreated} \neq \varnothing$ **do**

3: Select a node $i \in \Lambda_{untreated}$, then remove it from $\Lambda_{untreated}$ and add it to $\Lambda_{treated}$;

4: **For** label $L \in \Psi_{i}$ **do**

5: **For** feasible reachable node $j \in \Omega_{MC}\left(L\right)$ **do**

6: Extend label L to a new label L' with a node-based production plan $\gamma_{ih\tilde{h}td}^{n} \in \left\{\gamma_{ih\tilde{h}td}^{n} \in \Upsilon^{n} \mid a_{i,j,t+d} = 1\right\}$ (detailed in Appendix A.4), yielding a temporary set of labels *TempSet*

7: **If** $j \notin \Lambda_{treated}$ **do**

8: Add j to $\Lambda_{untreated}$, and update Ψ_{j} by sequentially adding label $L' \in TempSet$ if L' is not dominated by any label in Ψ_{j}

9: **Else**

10: Update UL by sequentially adding label $L' \in TempSet$ if L' is not dominated

by any label in UL

11: **While** $UL \neq \varnothing$ **do**

12: Select a label $L \in UL$, and remove it from UL

13: Implement the procedures as shown in Lines **5-10**

Output: The optimal migratory solution derived from the label associated with destination node e using forward dynamic programming

Algorithm A2 The LATD framework

1: Initialization: Label set $\Psi_{i_c t_c} = \{L\}$ at given initial state (i_c, t_c), label set $\Psi_{it} = \varnothing$ at node $i \in \mathcal{I} \cup \{e\}$ in period t, and set of nodes with untreated labels $\Lambda_t = \{i_c\}$ in period t

2: For $t \in \left[t_c, \max\left(t_c + \upsilon, |\mathcal{T}|\right)\right]$ **do**

3: **While** $\Lambda_t \neq \varnothing$ **do**

4: Select a node $i \in \Lambda_t$, then remove it from Λ_t;

5: **For** label $L \in \Psi_i$ **do**

6: **For** feasible reachable node $j \in \Omega_{TD}(L)$ **do**

7: Extend label L to new label L'

8: Add L' to $\Psi_{jt'}$ if L' is not dominated by any label in $\Psi_{jt'}$ and update $\Lambda_{t'} = \Lambda_{t'} \cup \{j\}$

9: **For** feasible production $(g, l) \in \mathcal{Z}$ **do**

10: Extend label L to new label L'

11: Add L' to $\Psi_{jt'}$ if L' is not dominated by any label in $\Psi_{jt'}$ and update $\Lambda_{t'} = \Lambda_{t'} \cup \{j\}$

12: Determine the returned label with the maximum profit among the labels satisfying $t(L) \geq \max\left(t_c + \upsilon, |\mathcal{T}|\right)$

Output: The optimal migratory solution derived from the returned label

A.4 The derivation of the node-based production plan

We formally introduce integrated production arrangements for each floral source, including production duration, product category, and production intensity. By identifying and eliminating dominant production decisions at each floral source, as stated in Definition A1, original periodic decisions are reformulated into a compact node-based planning framework, thereby improving computational efficiency.

Definition A1: Let $\Upsilon^n = \left(\gamma^n_{ihh'td}\right)_{\forall i \in \mathcal{I}, \forall h \in \mathcal{H}, \forall h' \in \Phi^n_{ihtd}, \forall t \in \left[T_i^S, T_i^E\right], \forall d \in \left[1, T_i^E - t\right]}$ denote the set of node-based production plans. Each plan $\gamma^n_{ihh'td}$ specifies that bee colonies carry out production at floral source i from period t to period $t+d$, during which colony strength transitions from h to h', yielding a total production revenue $\Pi\left(\gamma^n_{ihh'td}\right)$, where Φ^n_{ihtd} presents the attainable colony status within production periods. The detailed derivation is provided below.

Given an initial state $\left(y_{i\tau t}^{n},\zeta_{ht}^{n}\right)$, a production with d periods can be carried out, where $1\le d\le T_i^E-t$. When $d=1$, the feasible production decisions of the current period can be denoted as $\mathcal{Z}_{iht}^{n}=\left\{(g,l)\middle|\forall g\in\mathcal{G}_i,\forall l\in\mathcal{L},\beta_{ghlt}=1\right\}$. Once specific production $z_{iglt}^{n}\in\mathcal{Z}_{iht}^{n}$ is employed, we can obtain production yields $\alpha\psi p_g q_{ig}\eta_l z_{iglt}^{n}$ and resulting colony strength $h+w_{i,h,g,l,t+1}^{n}$. On this basis, the attainable colony strength incurred by production is $\Phi_{iht1}^{n}=\left\{h+w_{i,h,g,l,t+1}^{n}\,|\,(g,l)\in\mathcal{Z}_{iht}^{n}\right\}$. Therefore, when one-period production is carried out, for each $h'\in\Phi_{iht1}^{n}$, there is an optimal production plan $\gamma_{ihh't1}^{n}$ associated with the corresponding revenue $\Pi\left(\gamma_{ihh't1}^{n}\right)=\max\limits_{\forall\left\{(g,l)\in\mathcal{Z}_{iht}^{n}|h'=h+w_{ihglt}^{n}\right\}}\left(\alpha\psi p_g q_{ig}\eta_l z_{iglt}^{n}\right)$.

When increasing production periods to $d=2$, we can obtain production plans by iteratively traversing all feasible production operations of the current period $z_{iglt}^{n}\in\mathcal{Z}_{iht}^{n}$ under the initial state $\left(y_{i\tau t}^{n},\zeta_{ht}^{n}\right)$, and all feasible production operations $z_{i,g,l,t+1}^{n}\in\mathcal{Z}_{i,h_1,t+1}^{n}$ under the resulting new initial state $\left(y_{i,\tau,t+1}^{n},\zeta_{h_1,t+1}^{n}\right)$ in period $t+1$, where the new resulting colony strength depends on the initial system state and imposed production, i.e., $h_1=h+w_{i,h,g,l,t+1}^{n}$. After a two-period production, the attainable colony strengths are given by $\Phi_{iht2}^{n}=\left\{h'\,|\,\forall h_1\in\Phi_{iht1}^{n},\forall h'\in\Phi_{i,h_1,t+1,1}^{n}\right\}$. Therefore, when a two-period production is carried out, for each $h'\in\Phi_{iht2}^{n}$, there is an optimal production plan $\gamma_{ihh't2}^{n}$ associated with the corresponding revenue $\gamma_{ihh't2}^{n}=\max\limits_{\forall h_1\in\Phi_{iht1}}\left(\Pi\left(\gamma_{ihh_1t1}^{n}\right)+\Pi\left(\gamma_{i,h_1,h',t+1,1}^{n}\right)\right)$.

Similarly, when $d=m$, we can traverse all production plans $\gamma_{ihh_1td_1}^{n}\left|h_1\in\Phi_{ihtd_1}^{n}\right.$ with periods d_1 under the initial state $\left(y_{i\tau t}^{n},\zeta_{ht}^{n}\right)$ and production plans $\gamma_{i,h_1,h',t+d_1,m-d_1}^{n}$ with $m-d_1$ periods under the new initial state $\left(y_{i,\tau,t+d_1}^{n},\zeta_{h_1,t+d_1}^{n}\right)$, where d_1 takes any value between $[1,m-1]$. With a production spanning m periods, the resulting colony strengths are denoted as $\Phi_{ihtm}^{n}=\left\{h'\,|\,\forall h_1\in\Phi_{ihtd_1}^{n},\forall h'\in\Phi_{i,h_1,t+d_1,m-d_1}^{n}\right\}$. For each $h'\in\Phi_{ihtm}^{n}$, there is an optimal production plan $\gamma_{ihh'tm}^{n}$ associated with the corresponding revenue $\Pi\left(\gamma_{ihh'tm}^{n}\right)=\max\limits_{\forall h_1\in\Phi_{ihtd_1},\forall d_1\in[1,m-1]}\left(\Pi\left(\gamma_{ihh_1td_1}^{n}\right)+\Pi\left(\gamma_{i,h_1,h',t+d_1,m-d_1}^{n}\right)\right)$.

A.5 Key elements of LAMC

The components of the LAMC are elaborated below. Let a partial path from origin o be represented by a label $L=\left(s,\vec{p},R,\Omega\right)$, corresponding to (i) the current state $s=\left(i,t,h\right)$, including spatiotemporal position $\left(i,t\right)$ and colony strength h; (ii) the migration route $\vec{p}=\left\{o,1,...,i\right\}$; (iii) the accumulated revenue R up to reaching state s; and (iv) the set of reachable floral sources $\Omega_{MC}=\left\{j\in I\bigcup\left\{e\right\}\mid a_{ijt'}=1,j\notin\vec{p},\exists t'\in\left[t,T_i^E\right]\right\}$ from the current state s.

Based on the definition, label L can be extended by adding a node $j\in\Omega_{MC}\left(L\right)$. However, there are multiple optional production plans $\left\{\gamma_{ih\tilde{h}td}^n\in\Upsilon^n\mid a_{i,j,t+d}=1\right\}$. For each alternative plan $\gamma_{ih\tilde{h}td}^n$, label L can be extended to a new label L' according to the following functions: (i) $s'=\left(i',t',h'\right)=\left(j,\max\left(t+d+\kappa_{ij},T_j^S\right),\tilde{h}+\varphi_{i,j,\tilde{h},t+d}\right)$ signifies colony strength $\tilde{h}+\varphi_{i,j,\tilde{h},t+d}$ upon the start time of production $\max\left(t+d+\kappa_{ij},T_j^S\right)$ at node j; (ii) the updated migration route $\vec{p}'=\vec{p}\bigcup\left\{j\right\}$; (iii) the updated profit $R'=R+\Pi\left(\gamma_{ih\tilde{h}td}^n\right)-c_{ij}^T$; and (iv) $\Omega'_{MC}=\left\{k\in\mathcal{I}\bigcup\left\{e\right\}\mid a_{jkt'}=1,k\notin\vec{p}',\exists t'\in\left[t',T_j^E\right]\right\}$. Note that function (iii) is modified when $j=e$ by including the penalty cost, as $R'=R+\Pi\left(\gamma_{ih\tilde{h}td}^n\right)-c_{ij}^T-\max\left(\alpha c^H\left(h_0-\lfloor\lambda h'\rfloor\right),0\right)$.

The effectiveness of the labeling algorithm largely hinges on dominance rules, which prune dominated partial paths. Given label $L_1=\left(s_1,\vec{p}_1,R_1,\Omega_{MC}^1\right)$ dominates $L_2=\left(s_2,\vec{p}_2,R_2,\Omega_{MC}^2\right)$, three prerequisite conditions must be met, i.e., $i_1=i_2$, $t_1\le t_2$, and $\Omega_{MC}^1\supseteq\Omega_{MC}^2$, and one of the following conditions holds: (i) When $t_1=t_2$, both $h_1\ge h_2$ and $R_1\ge R_2$ are satisfied; (ii) When $t_1<t_2$, there exists a production plan $\gamma_{i_1,h_1,\tilde{h},t_1,t_2-t_1}^n\in\Upsilon^n$ such that the achieved revenue and attained colony strength are comparable, i.e., $\tilde{h}\ge h_2$ and $R_1+\Pi\left(\gamma_{i_1,h_1,\tilde{h},t_1,t_2-t_1}^n\right)\ge R_2$. In other words, when the labels occupy the same spatiotemporal position, or can achieve the spatiotemporal coincidence via production plans, the dominant label should outperform others in both accumulated revenue and colony strength.

A.6 Key elements of LATD

Let a partial path be represented by label $L=\left(s,R,\Omega_{TD},\mathcal{Z},s_{pre}\right)$, corresponding to (i) the

current state $s=(i,t,h)$, (ii) the accumulated revenue R, (iii) the set of reachable floral sources from the current state s, i.e., $\Omega_{TD}=\{j\in\mathcal{I}\cup\{e\}\mid a_{ijt}=1, j\notin\vec{p}\}$, (iv) the set of feasible production arrangements $\mathcal{Z}=\{(g,l)\mid\forall g\in\mathcal{G}_i,\forall l\in\mathcal{L},\beta_{ghlt}=1\}$, and (v) the previous state s_{pre} used to track the partial route from the origin to the current state.

Based on the definition, the extension becomes complicated by the distinction between migration decisions and production assignment, as well as the incorporation of approximation values.

In terms of migration, label L can be extended to label L' for each migration $j\in\Omega_{TD}$: (i) the updated state $s'=\left(j,\max\left(t+\kappa_{ij},T_j^S\right),h+\varphi_{ijht}\right)$; (ii) the revenue R' is updated by including migratory cost, penalty cost, and approximated value, with corresponding situation as shown in the following equation (A4); (iii) $\Omega'_{TD}=\{k\in\mathcal{I}\cup\{e\}\mid k\notin\vec{p}'\vee a_{jkt'}=1\}$; (iv) $\mathcal{Z}=\{(g,l)\mid\forall g\in\mathcal{G}_j,\forall l\in\mathcal{L},\beta_{g,h+\varphi_{ijht},l,t}=1\}$; and (v) $s_{pre}=(i,j,t)$.

$$R'=\begin{cases}R-c_{ij}^T+\max\left(\alpha c^H\left(h_0-\lfloor\lambda h'\rfloor\right),0\right), & \text{if } j=e\\ R-c_{ij}^T, & \text{if } j\neq e \text{ and } t'<t_c+\upsilon\\ R-c_{ij}^T+\lambda^{n-1}_{j,0,\ell\left(\hbar^k_{j,t'-1},h'\right),t'-1}, & \text{if } j\neq e \text{ and } t_c+\upsilon\le t'<|\mathcal{T}|\end{cases}\tag{A4}$$

In terms of production, label L can be extended to label L' for production $(g,l)\in\mathcal{Z}$: (i) if $t+1<t_c+\upsilon$ holds, then $s'=\left(i,t+1,h+w^n_{i,h,g,l,t+1}\right)$; otherwise, $s'=\left(i,t+1,h+\left\lfloor\sum_{\tilde{h}\in\mathcal{H}}\tilde{h}\mathrm{P}_{h\tilde{h}lg}\right\rfloor\right)$, incorporating the expected increment of colony strength under the imposed foraging intensity z^n_{iglt}; (ii) if $t+1<t_c+\upsilon$, then $R'=R+\alpha\psi p_g q_{ig}\eta_l z^n_{iglt}$; otherwise, $R'=R+\alpha\psi p_g q_{ig}\eta_l z^n_{iglt}+\lambda^{n-1}_{i0\ell\left(\hbar^k_{j,t},h'\right)t}$; (iii) $\Omega'_{TD}=\{k\in\mathcal{I}\cup\{e\}\mid k\notin\vec{p}'\vee a_{jkt'}=1\}$; (iv) $\mathcal{Z}=\{(g,l)\mid\forall g\in\mathcal{G}_i,\forall l\in\mathcal{L},\beta_{gh'lt'}=1\}$; and (v) $s_{pre}=s$.

Given two labels $L_1=\left((i_1,t_1,h_1),R_1,\Omega^1_{TD},\mathcal{Z}_1,s^1_{pre}\right)$ and $L_2=\left((i_2,t_2,h_2),R_2,\Omega^2_{TD},\mathcal{Z}_2,s^2_{pre}\right)$ with the same spatiotemporal position, i.e., $i_1=i_2$ and $t_1=t_2$, label L_1 dominates L_2 if the following conditions are satisfied: $\Omega^1_{TD}\supseteq\Omega^2_{TD}$, $h_1\ge h_2$, and $R_1\ge R_2$.

Appendix B: Detailed comparison between our ADP policy and benchmark policies

The comparison between our ADP policy and benchmarks is primarily based on revenue- and strength-based indicators, as illustrated in Fig. B1 and Fig**. B2**, respectively. Fig. B1 summarizes revenue-based results across three metrics: absolute revenue (Rev.), improvement percentage over practical baselines (Imp.), and optimality gap relative to the Posterior-RP policy (Gap), with four subgraphs corresponding to results for cases originating from four origins, respectively. The histogram shows revenues for all policies (left y-axis), while the line chart displays improvement percentages (right y-axis). Overall, ADP outperforms benchmarks across instance scales and different origins, ranked by Rolling-RP, Static-R/Greedy-P, and Static-R/Rule-based-P policies, with distinct improvement percentages ranging from 30.56% to 73.79%. Furthermore, the ADP solution closely approaches the posterior-optimal solutions with an average optimality gap of 1.25% to 4.87%.

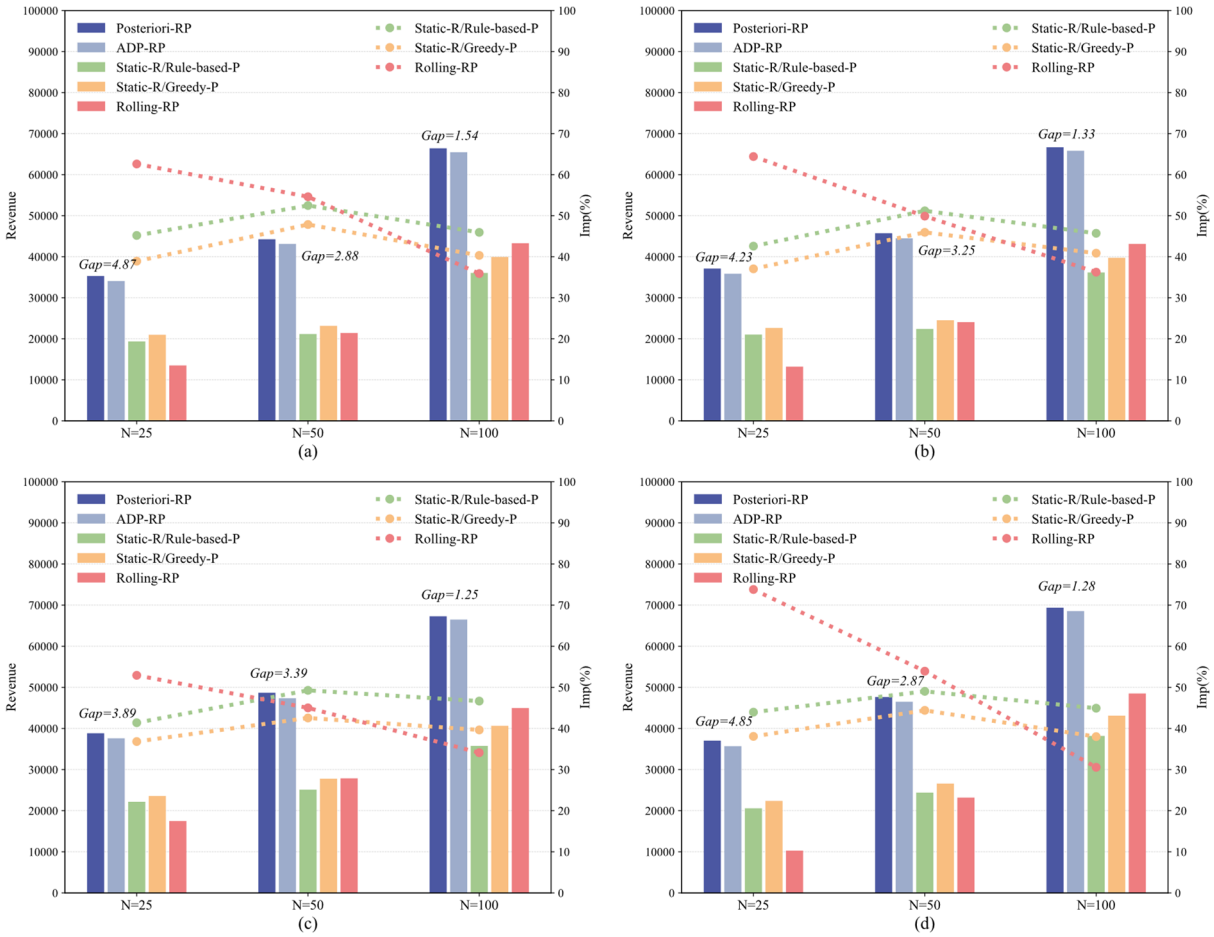

Fig. B1 Comparison of revenue-related metrics among different polices

Fig. B2 illustrates the distribution of colony strength under each policy, with a blue-to-red gradient indicating increasing levels of colony strength. The results show that ADP closely matches the posterior policy and maintains higher colony strength than other baselines, as

evidenced by fewer blue and more orange and red coverage. Specifically, the ADP policy sustains relatively good colony status across the majority of regions, approximately 60% to 70%, in orange and red. However, static-route-based policies yield a substantially lower proportion of moderate and robust colony status in orange and red regions across instance scales and different origins, i.e., 20% to 30% for Static-R/Rule-based-P, and only 10% to 20% for Static-R/Greedy-P. That is to say, colony status predominantly resides in a weak state over the time horizon. Rolling-RP presents different trends as the instance scale expands. Specifically, Rolling-RP achieves a pleasing colony status of 20%-30% for small-scale instances, roundly 60% for medium-scale instances, and even 70%-80% for large-scale instances.

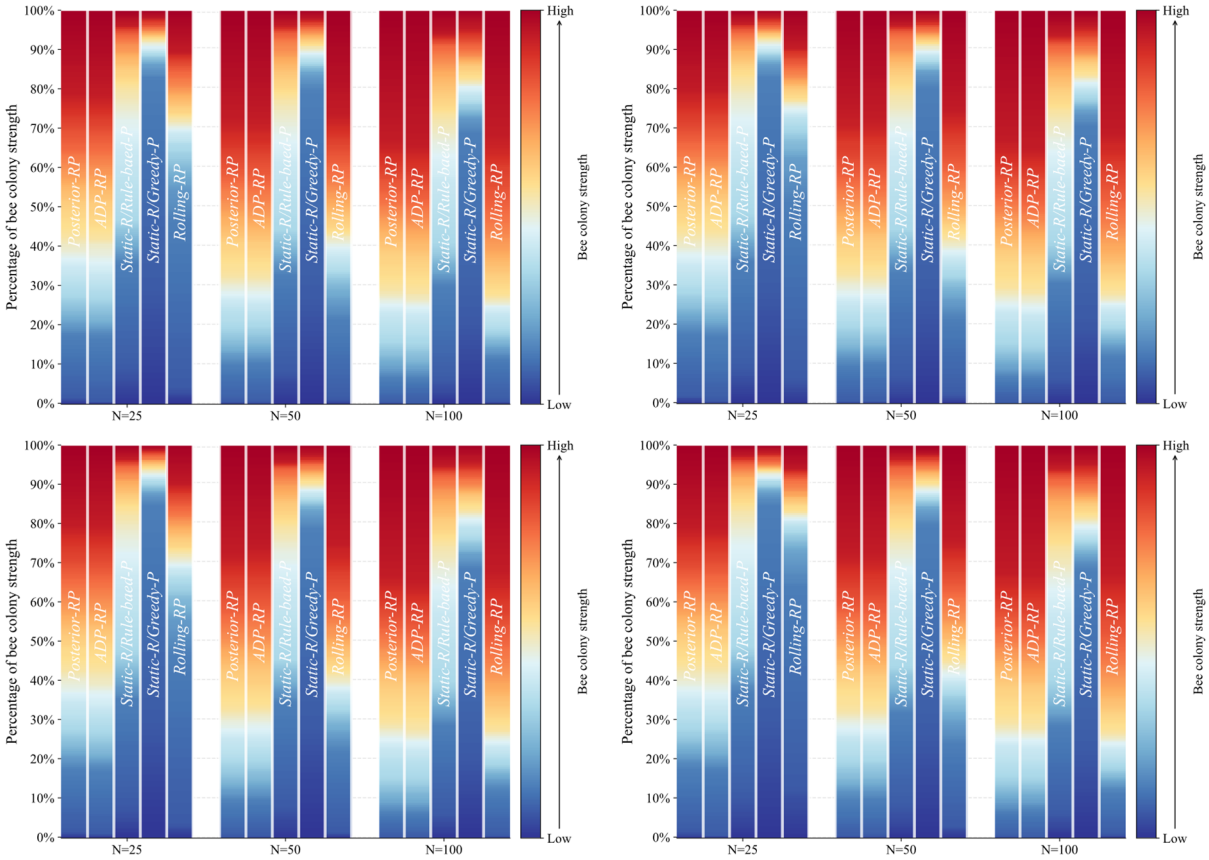

Fig. B2 The distribution of colony strength across different policies

Overall, substantial improvements over practical baselines and minor gaps relative to the posterior results hold consistently across instance scales and different origins, strongly validating the effectiveness and robustness of our ADP approach for the migratory beekeeping routing problem that considers stochastic transitions in colony status.

The above results reveal distinct differences in yields and colony status across different policies, stemming from underlying decision-making logic and their profound impacts on

yields and sustainability. The following analysis aims to explore the rationale behind and provide further insights. Given consistent trends in beekeeping revenue and colony strength across four origins, the results for the case originating from origin O1 are presented as an illustrative example. Table B1 provides a systematic comparison across multidimensional metrics, with the comparative results for cases originating from origins O2, O3, and O4 are presented in Supplement S4.2 of the online data repository.

Table B1 Comparison of KPIs across different policies based on instances originating from O1

N	Policy	Rev.	Imp. (%)	Gap (%)	Unit rev.	Unit income	Unit cost	#Source	Prod. period	Avg. str.	Avg. int.
25	Posterior-RP	35305.77	-	-	235.37	350.99	115.62	9.07	43.36	8.00	2.49
	ADP	34077.68	-	4.87	227.18	344.92	117.74	9.20	43.43	8.06	2.45
	Static-R/Rule-based-P	19345.99	45.19	-	128.97	238.14	109.16	8.80	43.10	4.38	1.90
	Static-R/Greedy-P	20982.87	38.91	-	139.89	249.05	109.16	8.80	43.10	1.73	1.95
	Rolling-RP	13498.24	62.61	-	89.99	168.92	78.93	6.17	28.09	4.13	2.09
50	Posterior-RP	44264.69	-	-	295.10	457.36	162.26	9.55	47.30	9.13	2.66
	ADP	43128.37	-	2.88	287.52	451.22	163.70	9.64	47.20	9.15	2.64
	Static-R/Rule-based-P	21167.06	52.48	-	141.11	288.73	147.61	8.70	46.90	4.82	1.94
	Static-R/Greedy-P	23156.37	47.85	-	154.38	301.99	147.61	8.70	46.90	1.92	1.98
	Rolling-RP	21416.4	54.63	-	142.78	267.15	124.37	7.41	33.12	7.33	2.37
100	Posterior-RP	66434.51	-	-	442.90	598.59	155.70	10.80	53.61	9.62	2.76
	ADP	65469.87	-	1.54	436.47	593.03	156.57	10.84	53.55	9.74	2.74
	Static-R/Rule-based-P	36040.61	45.94	-	240.27	378.81	138.54	9.80	53.40	5.32	2.00
	Static-R/Greedy-P	39943.15	40.32	-	266.29	404.83	138.54	9.80	53.40	2.99	2.08
	Rolling-RP	43298.44	35.89	-	288.66	428.28	139.62	10.00	43.83	9.51	2.58

Note: revenue (Unit rev.), gross income (Unit income) and expenditure (Unit cost) per bee colony, number of floral sources visited (#Source), average production periods (Prod. period), average colony strength (Avg. str.), and average production intensity level (Avg. int.)

Static-R/Greedy-P consistently yields higher revenue than Static-R/Rule-based-P, as indicated in column "Imp" with differences of 4.65% to 6.82%, despite the same static route. It is not surprising because Static-R/Rule-based-P implements production based on simple empirical rules rather than through the interaction between production operations and colony status transitions. In reality, colony strength after production operations varies with physical stress and workload, corresponding to distinct levels of productivity and resistance and, in turn, affecting subsequent profitability and sustainability. The greedy production decisions within predictable visions capture varying states to some extent and implement suitable production to improve overall revenue. It demonstrates the necessity of a comprehensive understanding of the transition of colony status and the fundamental role of production arrangements in ensuring that production operations align with the evolving colony status.

Compared with static-R-based policies, Rolling-RP integrates routing and production decisions, enabling it to perform well due to its enhanced flexibility. However, it performs inferiorly in small- and medium-scale instances, while showing anticipated advantages in large-scale scenarios. The trend can be explained as follows: Rolling-RP relies on myopic routing and production decisions within horizon visions, fails to identify profitable floral sources, and misses numerous production opportunities, as indicated in columns "#Source" and "Prod. period". However, static routes derived from expected conditions inherently incorporate future situations to some extent and might partially align with optimal routes. As the number of floral sources increases, Rolling-RP offers greater routing flexibility and provides a dynamic and suitable floral resource sequence, enhancing adaptability and achieving higher returns. Consequently, the critical role of routing decisions is evident. Without effective routing plans, it is difficult to achieve efficient production and superior revenue.

In contrast, the ADP policy offers the best achievable performance relative to the practical baselines and yields a win-win outcome between yields and colony status. The favorable result benefits from efficient routing scheduling with better temporal coherence and higher utilization of floral resources (columns "#Source" and "Prod. period"), and coupled with more reasonable production arrangements in compliance with involving complex interactions, which maintains a relatively robust colony status (column "Avg. str.") and supports further intensive production (column "Avg. int."). Fundamentally, the superior performance is attributed to jointly routing and production decisions, to the reliable approximation of system states, which lead to forward-looking, proactive decision-making over a long horizon, and to effective alignment between colonies and floral resources across spatial, temporal, and biomass aspects.